A Linear Algebra Review

In this appendix, we introduce some basic notions of linear algebra relevant to the material presented in this book. This appendix does not represent an exhaustive tutorial, and it is assumed that the reader has some prior knowledge of the subject.

A.1 Vectors and norms

We will denote by $\mathbb{H}$ a vector space whose dimension may be infinite.

A.1.1 Norms

Definition A.1 *A mapping $\Phi\colon \mathbb{H} \to \mathbb{R}_+$ is said to define a* norm *on $\mathbb{H}$ if it verifies the following axioms:*

- *definiteness:* $\forall \mathbf{x} \in \mathbb{H}, \Phi(\mathbf{x}) = 0 \Leftrightarrow \mathbf{x} = \mathbf{0}$*;*
- *homogeneity:* $\forall \mathbf{x} \in \mathbb{H}, \forall \alpha \in \mathbb{R}, \Phi(\alpha \mathbf{x}) = |\alpha| \Phi(\mathbf{x})$*;*
- *triangle inequality:* $\forall \mathbf{x}, \mathbf{y} \in \mathbb{H}, \Phi(\mathbf{x} + \mathbf{y}) \leq \Phi(\mathbf{x}) + \Phi(\mathbf{y})$.

A norm is typically denoted by $\|\cdot\|$. Examples of vector norms are the absolute value on $\mathbb{R}$ and the Euclidean (or L_2) norm on $\mathbb{R}^N$. More generally, for any $p \geq 1$ the L_p norm is defined on $\mathbb{R}^N$ as

$$\forall \mathbf{x} \in \mathbb{R}^N, \quad \|\mathbf{x}\|_p = \Big(\sum_{j=1}^{N} |x_j|^p \Big)^{1/p}. \tag{A.1}$$

The L_1, L_2, and L_∞ norms are some of the most commonly used norms, where $\|\mathbf{x}\|_\infty = \max_{j \in [N]} |x_j|$. Two norms $\|\cdot\|$ and $\|\cdot\|'$ are said to be *equivalent* iff there exists $\alpha, \beta > 0$ such that for all $\mathbf{x} \in \mathbb{H}$,

$$\alpha \|\mathbf{x}\| \leq \|\mathbf{x}\|' \leq \beta \|\mathbf{x}\|. \tag{A.2}$$

The following general inequalities relating these norms can be proven straightforwardly:

$$\|\mathbf{x}\|_2 \leq \|\mathbf{x}\|_1 \leq \sqrt{N} \|\mathbf{x}\|_2 \tag{A.3}$$

$$\|\mathbf{x}\|_\infty \leq \|\mathbf{x}\|_2 \leq \sqrt{N} \|\mathbf{x}\|_\infty \tag{A.4}$$

$$\|\mathbf{x}\|_\infty \leq \|\mathbf{x}\|_1 \leq N \|\mathbf{x}\|_\infty. \tag{A.5}$$

The second inequality of the first line can be shown using the *Cauchy-Schwarz inequality* presented later while the other inequalities are clear. These inequalities show the equivalence of these three norms. More generally, all norms on a finite-dimensional space are equivalent. The following additional properties hold for the L_∞ norm: for all $\mathbf{x} \in \mathbb{H}$,

$$\forall p \geq 1, \ \|\mathbf{x}\|_\infty \leq \|\mathbf{x}\|_p \leq N^{1/p} \|\mathbf{x}\|_\infty \tag{A.6}$$

$$\lim_{p \to +\infty} \|\mathbf{x}\|_p = \|\mathbf{x}\|_\infty. \tag{A.7}$$

The inequalities of the first line are straightforward and imply the limit property of the second line.

Definition A.2 (Hilbert space) *A* Hilbert space *is a vector space equipped with an inner product* $\langle \cdot, \cdot \rangle$ *and that is* complete *(all Cauchy sequences are convergent). The inner product induces a norm defined as follows:*

$$\forall \mathbf{x} \in \mathbb{H}, \quad \|\mathbf{x}\|_{\mathbb{H}} = \sqrt{\langle \mathbf{x}, \mathbf{x} \rangle}. \tag{A.8}$$

A.1.2 Dual norms

Definition A.3 *Let* $\|\cdot\|$ *be a norm on* $\mathbb{R}^N$*. Then, the* dual norm $\|\cdot\|_*$ *associated to* $\|\cdot\|$ *is the norm defined by*

$$\forall \mathbf{y} \in \mathbb{R}^N, \quad \|\mathbf{y}\|_* = \sup_{\|\mathbf{x}\|=1} |\langle \mathbf{y}, \mathbf{x} \rangle|. \tag{A.9}$$

For any $p, q \geq 1$ that are *conjugate* that is such that $\frac{1}{p} + \frac{1}{q} = 1$, the L_p and L_q norms are dual norms of each other. In particular, the dual norm of L_2 is the L_2 norm, and the dual norm of the L_1 norm is the L_∞ norm.

Proposition A.4 (Hölder's inequality) *Let* $p, q \geq 1$ *be conjugate:* $\frac{1}{p} + \frac{1}{q} = 1$*. Then, for all* $x, y \in \mathbb{R}^N$*,*

$$|\langle \mathbf{x}, \mathbf{y} \rangle| \leq \|\mathbf{x}\|_p \|\mathbf{y}\|_q, \tag{A.10}$$

with equality when $|y_i| = |x_i|^{p-1}$ *for all* $i \in [N]$*.*

Proof: The statement holds trivially for $\mathbf{x} = \mathbf{0}$ or $\mathbf{y} = \mathbf{0}$; thus, we can assume $\mathbf{x} \neq \mathbf{0}$ and $\mathbf{y} \neq \mathbf{0}$. Let $a, b > 0$. By the concavity of log (see definition B.7), we can write

$$\log\left(\frac{1}{p}a^p + \frac{1}{q}b^q\right) \geq \frac{1}{p}\log(a^p) + \frac{1}{q}\log(b^q) = \log(a) + \log(b) = \log(ab).$$

Taking the exponential of the left- and right-hand sides gives

$$\frac{1}{p}a^p + \frac{1}{q}b^q \geq ab,$$

which is known as *Young's inequality*. Using this inequality with $a = |x_j|/\|\mathbf{x}\|_p$ and $b = |y_j|/\|\mathbf{y}\|_q$ for $j \in [N]$ and summing up gives

$$\frac{\sum_{j=1}^N |x_j y_j|}{\|\mathbf{x}\|_p \|\mathbf{y}\|_q} \leq \frac{1}{p}\frac{\|\mathbf{x}\|^p}{\|\mathbf{x}\|^p} + \frac{1}{q}\frac{\|\mathbf{y}\|^q}{\|\mathbf{y}\|^q} = \frac{1}{p} + \frac{1}{q} = 1.$$

Since $|\langle \mathbf{x}, \mathbf{y} \rangle| \leq \sum_{j=1}^N |x_j y_j|$, the inequality claim follows. The equality case can be verified straightforwardly. □

Taking $p = q = 2$ immediately yields the following result known as the *Cauchy-Schwarz inequality*.

Corollary A.5 (Cauchy-Schwarz inequality) *For all* $\mathbf{x}, \mathbf{y} \in \mathbb{R}^N$*,*

$$|\langle \mathbf{x}, \mathbf{y} \rangle| \leq \|\mathbf{x}\|_2 \|\mathbf{y}\|_2, \tag{A.11}$$

with equality iff $\mathbf{x}$ *and* $\mathbf{y}$ *are collinear.*

Let $\mathcal{H}$ be the hyperplane in $\mathbb{R}^N$ whose equation is given by

$$\mathbf{w} \cdot \mathbf{x} + b = 0,$$

for some normal vector $\mathbf{w} \in \mathbb{R}^N$ and offset $b \in \mathbb{R}$. Let $d_p(\mathbf{x}, \mathcal{H})$ denote the distance of $\mathbf{x}$ to the hyperplane $\mathcal{H}$, that is,

$$d_p(\mathbf{x}, \mathcal{H}) = \inf_{\mathbf{x}' \in \mathcal{H}} \|\mathbf{x}' - \mathbf{x}\|_p. \tag{A.12}$$

Then, the following identity holds for all $p \geq 1$:

$$d_p(\mathbf{x}, \mathcal{H}) = \frac{|\mathbf{w} \cdot \mathbf{x} + b|}{\|\mathbf{w}\|_q}, \tag{A.13}$$

where q is the conjugate of p: $\frac{1}{p} + \frac{1}{q} = 1$. (A.13) can be shown by a straightforward application of the results of appendix B to the constrained optimization problem (A.12).

A.1.3 Relationship between norms

A general form for the inequalities seen in equations (A.3), (A.4) and (A.5), which holds for all L_p norms, is shown in the following proposition.

Proposition A.6 *Let $1 \leq p \leq q$. Then the following inequalities hold for all $\mathbf{x} \in \mathbb{R}^N$:*

$$\|x\|_q \leq \|x\|_p \leq N^{\frac{1}{p}-\frac{1}{q}} \|x\|_q \,. \tag{A.14}$$

Proof: First, assume $\mathbf{x} \neq \mathbf{0}$, otherwise the inequalities hold trivially. Then the first inequality holds using $1 \leq p \leq q$ as follows:

$$\left[\frac{\|\mathbf{x}\|_p}{\|\mathbf{x}\|_q}\right]^p = \sum_{i=1}^N \Bigg[\underbrace{\frac{x_i}{\|\mathbf{x}\|_q}}_{\leq 1}\Bigg]^p \geq \sum_{i=1}^N \left[\frac{x_i}{\|\mathbf{x}\|_q}\right]^q = 1.$$

Finally, the second inequality follows by using Hölder's inequality (proposition A.4)

$$\|\mathbf{x}\|_p = \left[\sum_{i=1}^N |x_i|^p\right]^{\frac{1}{p}} \leq \left[\left(\sum_{i=1}^N (|x_i|^p)^{\frac{q}{p}}\right)^{\frac{p}{q}} \left(\sum_{i=1}^N (1)^{\frac{q}{q-p}}\right)^{1-\frac{p}{q}}\right]^{\frac{1}{p}} = \|\mathbf{x}\|_q N^{\frac{1}{p}-\frac{1}{q}} \,,$$

which completes the proof. □

A.2 Matrices

For a matrix $\mathbf{M} \in \mathbb{R}^{m\times n}$ with m rows and n columns, we denote by $\mathbf{M}_{ij}$ its ijth entry, for all $i \in [m]$ and $j \in [n]$. For any $m \geq 1$, we denote by $\mathbf{I}_m$ the m-dimensional identity matrix, and refer to it as $\mathbf{I}$ when the dimension is clear from the context.

The *transpose* of $\mathbf{M}$ is denoted by $\mathbf{M}^\top$ and defined by $(\mathbf{M}^\top)_{ij} = \mathbf{M}_{ji}$ for all (i,j). For any two matrices $\mathbf{M} \in \mathbb{R}^{m\times n}$ and $\mathbf{N} \in \mathbb{R}^{n\times p}$, $(\mathbf{MN})^\top = \mathbf{N}^\top\mathbf{M}^\top$. $\mathbf{M}$ is said to be *symmetric* iff $\mathbf{M}_{ij} = \mathbf{M}_{ji}$ for all (i,j), that is, iff $\mathbf{M} = \mathbf{M}^\top$.

The *trace* of a square matrix $\mathbf{M}$ is denoted by $\mathrm{Tr}[\mathbf{M}]$ and defined as $\mathrm{Tr}[\mathbf{M}] = \sum_{i=1}^N \mathbf{M}_{ii}$. For any two matrices $\mathbf{M} \in \mathbb{R}^{m\times n}$ and $\mathbf{N} \in \mathbb{R}^{n\times m}$, the following identity holds: $\mathrm{Tr}[\mathbf{MN}] = \mathrm{Tr}[\mathbf{NM}]$. More generally, the following cyclic property holds with the appropriate dimensions for the matrices $\mathbf{M}$, $\mathbf{N}$, and $\mathbf{P}$:

$$\mathrm{Tr}[\mathbf{MNP}] = \mathrm{Tr}[\mathbf{PMN}] = \mathrm{Tr}[\mathbf{NPM}]. \tag{A.15}$$

The inverse of a square matrix $\mathbf{M}$, which exists when $\mathbf{M}$ has full rank, is denoted by $\mathbf{M}^{-1}$ and is the unique matrix satisfying $\mathbf{MM}^{-1} = \mathbf{M}^{-1}\mathbf{M} = \mathbf{I}$.

A.2.1 Matrix norms

A *matrix norm* is a norm defined over $\mathbb{R}^{m\times n}$ where m and n are the dimensions of the matrices considered. Many matrix norms, including those discussed below, satisfy the following *submultiplicative property*:

$$\|\mathbf{MN}\| \leq \|\mathbf{M}\|\|\mathbf{N}\|. \tag{A.16}$$

The *matrix norm induced* by the vector norm $\|\cdot\|_p$ or the *operator norm induced* by that norm is also denoted by $\|\cdot\|_p$ and defined by

$$\|\mathbf{M}\|_p = \sup_{\|\mathbf{x}\|_p \leq 1} \|\mathbf{Mx}\|_p \,. \tag{A.17}$$

The norm induced for $p = 2$ is known as the *spectral norm*, which equals the largest singular value of $\mathbf{M}$ (see section A.2.2), or the square-root of the largest eigenvalue of $\mathbf{M}^\top\mathbf{M}$:

$$\|\mathbf{M}\|_2 = \sigma_1(\mathbf{M}) = \sqrt{\lambda_{\max}(\mathbf{M}^\top\mathbf{M})}. \tag{A.18}$$

Not all matrix norms are induced by vector norms. The *Frobenius norm* denoted by $\|\cdot\|_F$ is the most notable of such norms and is defined by:

$$\|\mathbf{M}\|_F = \left(\sum_{i=1}^{m}\sum_{j=1}^{n}\mathbf{M}_{ij}^2\right)^{1/2}.$$

The Frobenius norm can be interpreted as the L_2 norm of a vector when treating $\mathbf{M}$ as a vector of size mn. It also coincides with the norm induced by the *Frobenius product*, which is the inner product defined for all $\mathbf{M}, \mathbf{N} \in \mathbb{R}^{m\times n}$ by

$$\langle \mathbf{M}, \mathbf{N}\rangle_F = \mathrm{Tr}[\mathbf{M}^\top \mathbf{N}]. \tag{A.19}$$

This relates the Frobenius norm to the singular values of $\mathbf{M}$:

$$\|\mathbf{M}\|_F^2 = \mathrm{Tr}[\mathbf{M}^\top \mathbf{M}] = \sum_{i=1}^{r}\sigma_i(\mathbf{M})^2\,,$$

where $r = \mathrm{rank}(\mathbf{M})$. The second equality follows from properties of SPSD matrices (see section A.2.3).

For any $j \in [n]$, let $\mathbf{M}_j$ denote the jth column of $\mathbf{M}$, that is $\mathbf{M} = [\mathbf{M}_1 \cdots \mathbf{M}_n]$. Then, for any $p, r \geq 1$, the $L_{p,r}$ *group norm* of $\mathbf{M}$ is defined by

$$\|\mathbf{M}\|_{p,r} = \left(\sum_{j=1}^{n}\|\mathbf{M}_i\|_p^r\right)^{1/r}.$$

One of the most commonly used group norms is the $L_{2,1}$ norm defined by

$$\|\mathbf{M}\|_{2,1} = \sum_{i=1}^{n}\|\mathbf{M}_i\|_2\,.$$

A.2.2 Singular value decomposition

The compact *singular value decomposition (SVD)* of $\mathbf{M}$, with $r = \mathrm{rank}(\mathbf{M}) \leq \min(m, n)$, can be written as follows:

$$\mathbf{M} = \mathbf{U}_M \mathbf{\Sigma}_M \mathbf{V}_M^\top\,.$$

The $r \times r$ matrix $\mathbf{\Sigma}_M = \mathrm{diag}(\sigma_1, \ldots, \sigma_r)$ is diagonal and contains the non-zero *singular values* of $\mathbf{M}$ sorted in decreasing order, that is $\sigma_1 \geq \ldots \geq \sigma_r > 0$. The matrices $\mathbf{U}_M \in \mathbb{R}^{m\times r}$ and $\mathbf{V}_M \in \mathbb{R}^{n\times r}$ have orthonormal columns that contain the *left* and *right singular vectors* of $\mathbf{M}$ corresponding to the sorted singular values. We denote by $\mathbf{U}_k \in \mathbb{R}^{m\times k}$ the top $k \leq r$ left singular vectors of $\mathbf{M}$.

The *orthogonal projection* onto the span of $\mathbf{U}_k$ can be written as $\mathbf{P}_{U_k} = \mathbf{U}_k\mathbf{U}_k^\top$, where $\mathbf{P}_{U_k}$ is SPSD and idempotent, i.e., $\mathbf{P}_{U_k}^2 = \mathbf{P}_{U_k}$. Moreover, the orthogonal projection onto the subspace orthogonal to $\mathbf{U}_k$ is defined as $\mathbf{P}_{U_k,\perp}$. Similar definitions, i.e., $\mathbf{V}_k, \mathbf{P}_{V_k}, \mathbf{P}_{V_k,\perp}$, hold for the right singular vectors.

The *generalized inverse*, or *Moore-Penrose pseudo-inverse* of a matrix $\mathbf{M}$ is denoted by $\mathbf{M}^\dagger$ and defined by

$$\mathbf{M}^\dagger = \mathbf{U}_M \mathbf{\Sigma}_M^\dagger \mathbf{V}_M^\top, \tag{A.20}$$

where $\mathbf{\Sigma}_M^\dagger = \mathrm{diag}(\sigma_1^{-1}, \ldots, \sigma_r^{-1})$. For any square $m \times m$ matrix $\mathbf{M}$ with full rank, i.e., $r = m$, the pseudo-inverse coincides with the matrix inverse: $\mathbf{M}^\dagger = \mathbf{M}^{-1}$.

A.2.3 Symmetric positive semidefinite (SPSD) matrices

Definition A.7 *A symmetric matrix* $\mathbf{M} \in \mathbb{R}^{m\times m}$ *is said to be* positive semidefinite *iff*

$$\mathbf{x}^\top \mathbf{M}\mathbf{x} \geq 0 \tag{A.21}$$

for all $\mathbf{x} \in \mathbb{R}^m$. $\mathbf{M}$ *is said to be* positive definite *if the inequality is strict.*

Kernel matrices (see chapter 6) and orthogonal projection matrices are two examples of SPSD matrices. It is straightforward to show that a matrix $\mathbf{M}$ is SPSD iff its eigenvalues are all non-negative. Furthermore, the following properties hold for any SPSD matrix $\mathbf{M}$:

- $\mathbf{M}$ admits a decomposition $\mathbf{M} = \mathbf{X}^\top \mathbf{X}$ for some matrix $\mathbf{X}$ and the *Cholesky decomposition* provides one such decomposition in which $\mathbf{X}$ is an upper triangular matrix.
- The left and right singular vectors of $\mathbf{M}$ are the same and the SVD of $\mathbf{M}$ is also its eigenvalue decomposition.
- The SVD of an arbitrary matrix $\mathbf{X} = \mathbf{U}_X \mathbf{\Sigma}_X \mathbf{V}_X^\top$ defines the SVD of two related SPSD matrices: the left singular vectors ($\mathbf{U}_X$) are the eigenvectors of $\mathbf{X}\mathbf{X}^\top$, the right singular vectors ($\mathbf{V}_X$) are the eigenvectors of $\mathbf{X}^\top\mathbf{X}$ and the non-zero singular values of $\mathbf{X}$ are the square roots of the non-zero eigenvalues of $\mathbf{X}\mathbf{X}^\top$ and $\mathbf{X}^\top\mathbf{X}$.
- The trace of $\mathbf{M}$ is the sum of its singular values, i.e., $\mathrm{Tr}[\mathbf{M}] = \sum_{i=1}^{r} \sigma_i(\mathbf{M})$, where $\mathrm{rank}(\mathbf{M}) = r$.
- The top singular vector of $\mathbf{M}$, $\mathbf{u}_1$, maximizes the *Rayleigh quotient*, which is defined as
$$r(\mathbf{x}, \mathbf{M}) = \frac{\mathbf{x}^\top \mathbf{M} \mathbf{x}}{\mathbf{x}^\top \mathbf{x}} .$$
In other words, $\mathbf{u}_1 = \mathrm{argmax}_{\mathbf{x}}\, r(\mathbf{x}, \mathbf{M})$ and $r(\mathbf{u}, \mathbf{M}) = \sigma_1(\mathbf{M})$. Similarly, if $\mathbf{M}' = \mathbf{P}_{U_i,\perp}\mathbf{M}$, that is, the projection of $\mathbf{M}$ onto the subspace orthogonal to $\mathbf{U}_i$, then $\mathbf{u}_{i+1} = \mathrm{argmax}_{\mathbf{x}}\, r(\mathbf{x}, \mathbf{M}')$, where $\mathbf{u}_{i+1}$ is the $(i+1)$st singular vector of $\mathbf{M}$.